

CUSTOMER MILESTONE BRIEF

MS #47 - Graph-Bound Live Evidence Control Tower

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.47.0, 0.47.1, 0.47.2, 0.47.3
Date	2026-09-01
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #47 - Graph-Bound Live Evidence Control Tower converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Gives customers, operators, and executives a credible live command-post view whose status is visibly bound to the same Knowledge Graph identity used by the underlying evidence.

Why this matters to a customer

Gives customers, operators, and executives a credible live command-post view whose status is visibly bound to the same Knowledge Graph identity used by the underlying evidence.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- An approved warm editorial product shell using paper-white work surfaces, quiet stone rails, burnt-orange hierarchy, locally served IBM Plex typography, sentence case, rounded controls, and visible interaction states.
- A graph-bound Company to Business problem to Workload to Technology scope to Operator lens path with session-stable estate context separated from active work controls.
- A dedicated Knowledge Graph binding card showing canonical content identity, entity and relationship counts, freshness, and downstream binding status.
- A labelled claim band showing one snapshot identity plus explicit entity, relationship, rule, and validated-extension values.
- A 70-node readability guard whose reduction and full-render actions dismiss correctly, with package-collapse, rules-and-proof, and legacy-to-modern choices.
- Collision-aware graph labels across five candidate positions, with tooltip preservation and suppression only when no readable placement exists.
- Keyboard-operable graph nodes with roving focus, spatial arrow-key navigation, Enter/Space inspection, Home-to-root behavior, descriptive names, and visible focus.

- Readable top-control and repeated metadata labels plus a non-clipping accessible live connection status.
- A proof-oriented workload action that opens the existing governed proof-run evidence from the inspector.
- A first-class graph operational source with hash-chained change events, five-second freshness refresh, and fail-closed downstream identity validation.
- Default refusal of non-loopback binds, an explicit exposure override, and a generated per-session bearer token required for verifier-audience HTTP routes.
- Locally served SIL-OFL IBM Plex webfonts with no Google Fonts or other runtime asset request.
- Regression coverage across policy, HTTP authorization, UI contracts, graph binding, density behavior, label placement, freshness, and documentation.

How it differs from earlier work

MS #17 introduced the live read-only evidence plane, MS #46 added unified-estate navigation and trace, and MS #47 makes the canonical graph itself observable. MS #47.1 through #47.3 close the independent visual, usability, density, local-transport, craft, and accessibility reviews without weakening the evidence boundary.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use graph-bound live evidence control tower as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- The Control Tower remains a read-only projection and does not gain approve, dispatch, retry, promote, or exception-authoring authority.
- Implementer routes remain unauthenticated and are safe only on loopback; the explicit non-loopback override does not make a deployment secure.
- The verifier token is per-session audience authorization, not enterprise identity, RBAC, or a substitute for customer SSO/OIDC at a reverse proxy.
- Graph freshness proves observation of the committed snapshot and evidence pack; it does not prove customer runtime behavior.

- A graph change invalidates mismatched downstream evidence; it does not automatically regenerate or approve that evidence.
- Customer execution evidence, native platform equivalence, and production readiness remain false.

Source of record

- CHANGELOG.md release section(s): 0.47.0, 0.47.1, 0.47.2, 0.47.3
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity